Outlook

**Oil forecast 2026-10-25: brent_fut 110.40 | XGBoost +60d 110.2**

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# Oil forecast report

## Signal summary

### Brent futures: next 1 trading day

Consensus: **NEUTRAL** (0 BUY, 1 SELL, 4 HOLD) | best track record among the agreeing models: **-**

| Model | Signal | Expected move | Move / noise (sd) | Model-implied P(up) | Direction hit-rate in frozen test (effective n) | Track record |
|---|---|---|---|---|---|---|
| XGBoost | **SELL** | -0.25% | 0.10 | 46% | 54% (n=451) | **Suggestive** |
| ARMA-GARCH | **HOLD** | +0.10% | 0.04 | 51% | 51% (n=451) | **None yet** |
| Regime LASSO | **HOLD** | -0.05% | 0.02 | 49% | 50% (n=451) | **None yet** |
| MS-AR | **HOLD** | -0.22% | 0.06 | 48% | 48% (n=451) | **None yet** |
| MS-ARX | **HOLD** | +0.16% | 0.04 | 52% | 46% (n=451) | **None yet** |

### WTI futures: next 1 trading day

Consensus: **NEUTRAL** (0 BUY, 0 SELL, 5 HOLD) | best track record among the agreeing models: **-**

| Model | Signal | Expected move | Move / noise (sd) | Model-implied P(up) | Direction hit-rate in frozen test (effective n) | Track record |
|---|---|---|---|---|---|---|
| XGBoost | **HOLD** | -0.00% | 0.00 | 50% | 51% (n=451) | **None yet** |
| ARMA-GARCH | **HOLD** | -0.08% | 0.03 | 49% | 51% (n=451) | **None yet** |
| Regime LASSO | **HOLD** | +0.00% | 0.00 | 50% | 49% (n=451) | **None yet** |
| MS-AR | **HOLD** | -0.07% | 0.02 | 49% | 48% (n=451) | **None yet** |
| MS-ARX | **HOLD** | +0.25% | 0.06 | 53% | 47% (n=451) | **None yet** |

HOLD = the expected move is smaller than 0.1 x its 1-sd noise, so there is nothing to act on. **Track record** = how the model's direction did in the frozen test (is it right significantly more often than a coin flip, and does it beat "no change"?): *Significant* = yes on both; *Suggestive* = some evidence ($p < 0.10$, or a hit-rate above 55% with a better error); *None yet* = no evidence of an edge; *No call* = the model forecast exactly no change. Rows are ordered by track record, strongest first.

# Brent futures (brent_fut) as of 2026-10-25: **110.40** USD/bbl

## Forecasts from today's data

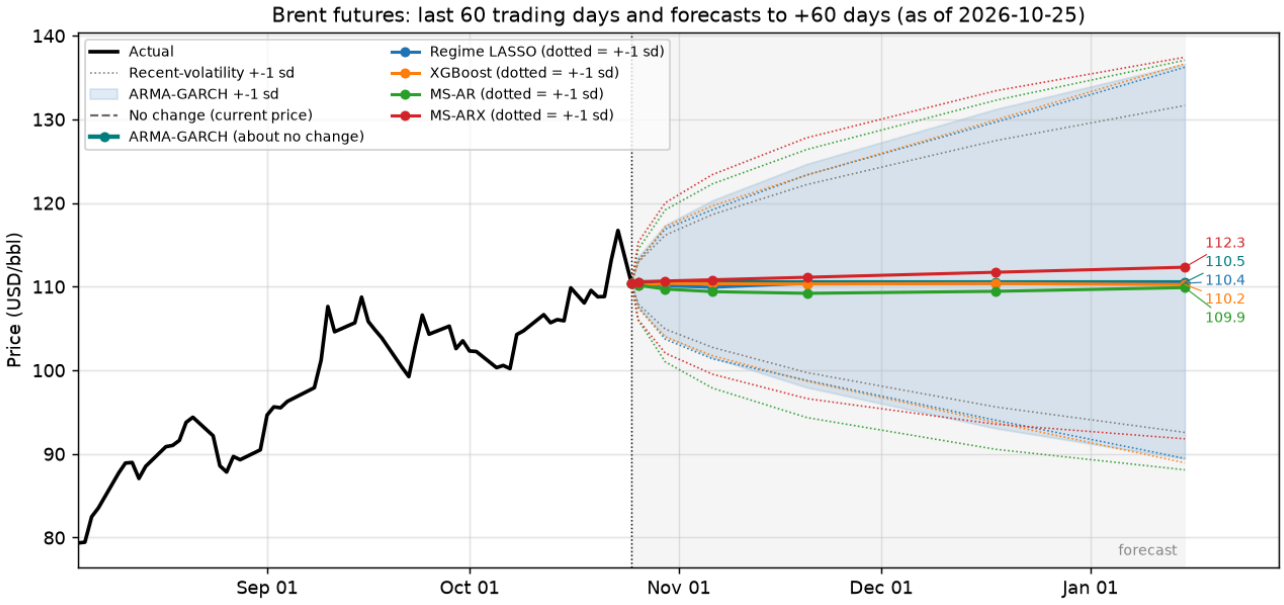
Price (change vs no change). Grey = same as no change.

| Horizon | Target date | No change | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX |
|---|---|---|---|---|---|---|---|
| +1d | 2026-10-26 | 110.40 | 110.34 (-0.1%) | 110.12 (-0.3%) | 110.51 (+0.1%) | 110.16 (-0.2%) | 110.58 (+0.2%) |
| +5d | 2026-10-30 | 110.40 | 110.11 (-0.3%) | 110.39 (-0.0%) | 110.51 (+0.1%) | 109.72 (-0.6%) | 110.68 (+0.3%) |
| +10d | 2026-11-06 | 110.40 | 109.93 (-0.4%) | 110.35 (-0.0%) | 110.51 (+0.1%) | 109.40 (-0.9%) | 110.83 (+0.4%) |
| +20d | 2026-11-20 | 110.40 | 110.40 (+0.0%) | 110.32 (-0.1%) | 110.51 (+0.1%) | 109.21 (-1.1%) | 111.12 (+0.7%) |
| +40d | 2026-12-18 | 110.40 | 110.43 (+0.0%) | 110.38 (-0.0%) | 110.51 (+0.1%) | 109.45 (-0.9%) | 111.72 (+1.2%) |
| +60d | 2027-01-15 | 110.40 | 110.40 (+0.0%) | 110.25 (-0.1%) | 110.51 (+0.1%) | 109.89 (-0.5%) | 112.33 (+1.7%) |

## Uncertainty: +-1 sd range of the price

| Horizon | Recent volatility (20d) | Regime LASSO (model) | XGBoost (model) | ARMA-GARCH (model) | MS-AR (model) | MS-ARX (model) |
|---|---|---|---|---|---|---|
| +1d | 107.9 - 112.9 (2.3%) | 107.6 - 113.2 (2.5%, x1.12) | 107.4 - 112.9 (2.5%, x1.10) | 107.5 - 113.6 (2.7%, x1.20) | 106.0 - 114.5 (3.8%, x1.69) | 106.0 - 115.3 (4.2%, x1.84) |
| +5d | 104.9 - 116.2 (5.1%) | 103.7 - 116.9 (6.0%, x1.17) | 104.0 - 117.2 (6.0%, x1.18) | 104.1 - 117.4 (6.0%, x1.18) | 101.0 - 119.2 (8.3%, x1.63) | 102.1 - 120.0 (8.1%, x1.59) |
| +10d | 102.7 - 118.6 (7.2%) | 101.4 - 119.2 (8.1%, x1.13) | 101.8 - 119.7 (8.1%, x1.13) | 101.5 - 120.3 (8.5%, x1.18) | 97.9 - 122.3 (11.1%, x1.55) | 99.5 - 123.4 (10.7%, x1.50) |
| +20d | 99.7 - 122.2 (10.2%) | 98.8 - 123.4 (11.1%, x1.09) | 98.7 - 123.3 (11.1%, x1.10) | 98.0 - 124.7 (12.1%, x1.19) | 94.3 - 126.4 (14.6%, x1.44) | 96.6 - 127.8 (14.0%, x1.38) |
| +40d | 95.6 - 127.5 (14.4%) | 94.0 - 129.7 (16.1%, x1.12) | 93.8 - 129.9 (16.3%, x1.14) | 93.1 - 131.2 (17.2%, x1.19) | 90.6 - 132.3 (19.0%, x1.32) | 93.6 - 133.4 (17.7%, x1.23) |
| +60d | 92.6 - 131.7 (17.6%) | 89.5 - 136.2 (21.0%, x1.19) | 88.9 - 136.7 (21.5%, x1.22) | 89.4 - 136.6 (21.2%, x1.20) | 88.1 - 137.1 (22.1%, x1.26) | 91.8 - 137.4 (20.2%, x1.15) |

## Last 60 trading days and forecasts




## Last 7 days: actual vs the 1-day-ahead forecast made the day before

| Date | Actual | No change | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX |
|---|---|---|---|---|---|---|---|
| 2026-10-25 | 110.40 | 116.72 (-5.4%) | 116.68 (-5.4%) | 116.56 (-5.3%) | 116.65 (-5.4%) | 116.83 (-5.5%) | 116.85 (-5.5%) |
| 2026-10-23 | 116.72 | 112.79 (+3.5%) | 112.75 (+3.5%) | 112.77 (+3.5%) | 112.71 (+3.6%) | 112.92 (+3.4%) | 112.79 (+3.5%) |
| 2026-10-22 | 113.17 | 110.04 (+2.8%) | 109.99 (+2.9%) | 110.03 (+2.9%) | 110.02 (+2.9%) | 110.13 (+2.8%) | 109.90 (+3.0%) |
| 2026-10-21 | 108.82 | 109.40 (-0.5%) | 109.35 (-0.5%) | 109.51 (-0.6%) | 109.40 (-0.5%) | 109.47 (-0.6%) | 109.27 (-0.4%) |
| 2026-10-20 | 108.79 | 109.88 (-1.0%) | 109.84 (-1.0%) | 109.99 (-1.1%) | 109.85 (-1.0%) | 109.98 (-1.1%) | 109.90 (-1.0%) |
| 2026-10-19 | 109.55 | 108.05 (+1.4%) | 108.05 (+1.4%) | 108.12 (+1.3%) | 108.08 (+1.4%) | 108.08 (+1.4%) | 108.01 (+1.4%) |
| 2026-10-18 | 108.05 | 109.86 (-1.6%) | 109.86 (-1.6%) | 109.87 (-1.7%) | 109.79 (-1.6%) | 109.98 (-1.8%) | 109.84 (-1.6%) |

Cell = forecast made the previous trading day (actual / forecast - 1 in brackets).

## Track record: RMSE vs no change (below 1.000 = better than no change)

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | n |
|---|---|---|---|---|---|---|
| +1d | 1.000 | 1.038 | 1.018 | 1.009 | 1.023 | 71 |
| +5d | 0.999 | 0.874 | 1.082 | 1.039 | 1.077 | 67 |
| +10d | 1.000 | 0.770 | 1.154 | 1.098 | 1.148 | 58 |
| +20d | 1.000 | 0.850 | 1.098 | 1.064 | 1.097 | 38 |
| +40d | n/a | n/a | n/a | n/a | n/a | 2 |

Share of outcomes inside the +-1 sd band (about 68% if the band is well calibrated):

| Horizon | Recent volatility | XGBoost | ARMA-GARCH | MS-AR | MS-ARX |
|---|---|---|---|---|---|
| +1d | 66% | 63% | 73% | 73% | 76% |
| +5d | 76% | 90% | 70% | 68% | 62% |
| +10d | 78% | 96% | 75% | 69% | 66% |
| +20d | 95% | 100% | 91% | 91% | 91% |

| | | | | | |
|---|---|---|---|---|---|
| +40d | n/a | n/a | n/a | n/a | n/a |

# WTI futures (wti_fut) as of 2026-10-25: **101.74** USD/bbl

## Forecasts from today's data

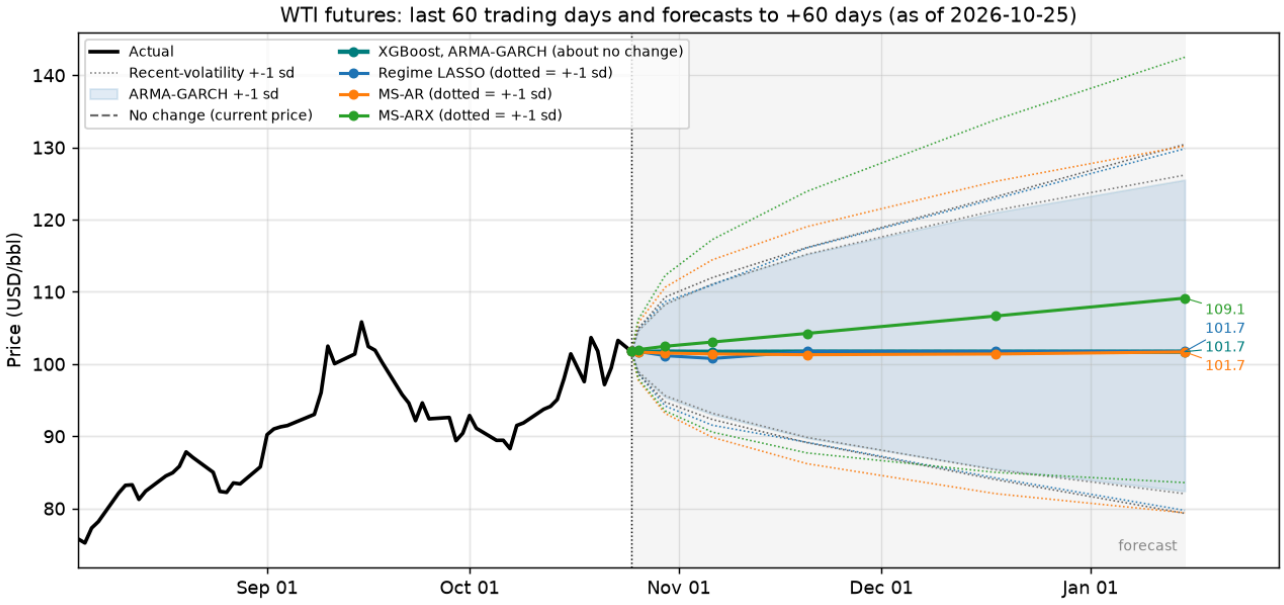
Price (change vs no change). Grey = same as no change.

| Horizon | Target date | No change | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX |
|---|---|---|---|---|---|---|---|
| +1d | 2026-10-26 | 101.74 | 101.74 (+0.0%) | 101.74 (-0.0%) | 101.66 (-0.1%) | 101.66 (-0.1%) | 102.00 (+0.3%) |
| +5d | 2026-10-30 | 101.74 | 101.16 (-0.6%) | 101.74 (+0.0%) | 101.70 (-0.0%) | 101.52 (-0.2%) | 102.46 (+0.7%) |
| +10d | 2026-11-06 | 101.74 | 100.77 (-1.0%) | 101.67 (-0.1%) | 101.70 (-0.0%) | 101.40 (-0.3%) | 103.05 (+1.3%) |
| +20d | 2026-11-20 | 101.74 | 101.74 (+0.0%) | 101.74 (-0.0%) | 101.70 (-0.0%) | 101.29 (-0.4%) | 104.24 (+2.5%) |
| +40d | 2026-12-18 | 101.74 | 101.74 (+0.0%) | 101.72 (-0.0%) | 101.70 (-0.0%) | 101.39 (-0.3%) | 106.66 (+4.8%) |
| +60d | 2027-01-15 | 101.74 | 101.74 (+0.0%) | 101.72 (-0.0%) | 101.70 (-0.0%) | 101.68 (-0.1%) | 109.13 (+7.3%) |

## Uncertainty: +-1 sd range of the price

| Horizon | Recent volatility (20d) | Regime LASSO (model) | XGBoost (model) | ARMA-GARCH (model) | MS-AR (model) | MS-ARX (model) |
|---|---|---|---|---|---|---|
| +1d | 99.0 - 104.6 (2.8%) | 98.7 - 104.9 (3.1%, x1.10) | 98.7 - 104.9 (3.1%, x1.10) | 98.7 - 104.7 (3.0%, x1.06) | 97.8 - 105.7 (3.9%, x1.40) | 98.0 - 106.2 (4.0%, x1.44) |
| +5d | 95.6 - 108.3 (6.2%) | 94.2 - 108.6 (7.1%, x1.15) | 94.7 - 109.3 (7.1%, x1.15) | 95.4 - 108.4 (6.4%, x1.03) | 93.2 - 110.6 (8.6%, x1.38) | 93.5 - 112.3 (9.2%, x1.47) |
| +10d | 93.2 - 111.1 (8.8%) | 91.5 - 111.0 (9.6%, x1.10) | 92.3 - 112.0 (9.7%, x1.10) | 93.0 - 111.2 (8.9%, x1.02) | 89.8 - 114.4 (12.1%, x1.38) | 90.6 - 117.3 (12.9%, x1.47) |
| +20d | 89.9 - 115.2 (12.4%) | 89.2 - 116.1 (13.2%, x1.06) | 89.1 - 116.1 (13.2%, x1.07) | 89.8 - 115.2 (12.5%, x1.01) | 86.2 - 119.0 (16.1%, x1.30) | 87.7 - 123.9 (17.3%, x1.39) |
| +40d | 85.3 - 121.3 (17.6%) | 84.2 - 122.9 (18.9%, x1.08) | 84.0 - 123.2 (19.1%, x1.09) | 85.5 - 121.0 (17.4%, x0.99) | 82.0 - 125.3 (21.2%, x1.21) | 85.0 - 133.8 (22.7%, x1.29) |
| +60d | 82.0 - 126.2 (21.5%) | 79.7 - 129.8 (24.4%, x1.13) | 79.3 - 130.4 (24.9%, x1.16) | 82.4 - 125.5 (21.0%, x0.98) | 79.4 - 130.2 (24.7%, x1.15) | 83.6 - 142.5 (26.7%, x1.24) |

## Last 60 trading days and forecasts




## Last 7 days: actual vs the 1-day-ahead forecast made the day before

| Date | Actual | No change | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX |
|---|---|---|---|---|---|---|---|
| 2026-10-25 | 101.74 | 103.27 (-1.5%) | 103.27 (-1.5%) | 103.28 (-1.5%) | 103.15 (-1.4%) | 103.30 (-1.5%) | 103.46 (-1.7%) |
| 2026-10-23 | 103.27 | 99.14 (+4.2%) | 99.14 (+4.2%) | 99.14 (+4.2%) | 99.23 (+4.1%) | 99.14 (+4.2%) | 99.29 (+4.0%) |
| 2026-10-22 | 99.47 | 98.07 (+1.4%) | 98.07 (+1.4%) | 98.07 (+1.4%) | 98.18 (+1.3%) | 97.95 (+1.6%) | 98.17 (+1.3%) |
| 2026-10-21 | 97.15 | 98.42 (-1.3%) | 98.42 (-1.3%) | 98.42 (-1.3%) | 98.34 (-1.2%) | 98.24 (-1.1%) | 98.52 (-1.4%) |
| 2026-10-20 | 101.79 | 98.80 (+3.0%) | 98.80 (+3.0%) | 98.81 (+3.0%) | 98.88 (+2.9%) | 98.86 (+3.0%) | 98.93 (+2.9%) |
| 2026-10-19 | 103.66 | 97.56 (+6.3%) | 97.56 (+6.3%) | 97.57 (+6.2%) | 97.53 (+6.3%) | 97.50 (+6.3%) | 97.71 (+6.1%) |
| 2026-10-18 | 97.56 | 99.02 (-1.5%) | 99.02 (-1.5%) | 101.41 (-3.8%) | 98.92 (-1.4%) | 99.08 (-1.5%) | 99.09 (-1.5%) |

Cell = forecast made the previous trading day (actual / forecast - 1 in brackets).

## Track record: RMSE vs no change (below 1.000 = better than no change)

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | n |
|---|---|---|---|---|---|---|
| +1d | 1.000 | 0.957 | 0.931 | 0.946 | 0.943 | 41 |
| +5d | 0.995 | 1.016 | 1.134 | 0.995 | 0.975 | 37 |
| +10d | 0.988 | 1.000 | 1.158 | 0.975 | 0.909 | 32 |
| +20d | 1.000 | 1.000 | 0.894 | 1.000 | 0.853 | 22 |
| +40d | n/a | n/a | n/a | n/a | n/a | 2 |

Share of outcomes inside the +-1 sd band (about 68% if the band is well calibrated):

| Horizon | Recent volatility | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX |
|---|---|---|---|---|---|---|
| +1d | 57% | 59% | 63% | 63% | 66% | 69% |
| +5d | 58% | 64% | 65% | 51% | 65% | 68% |
| +10d | 50% | 69% | 69% | 47% | 65% | 85% |
| +20d | 88% | 100% | 100% | 95% | 94% | 100% |

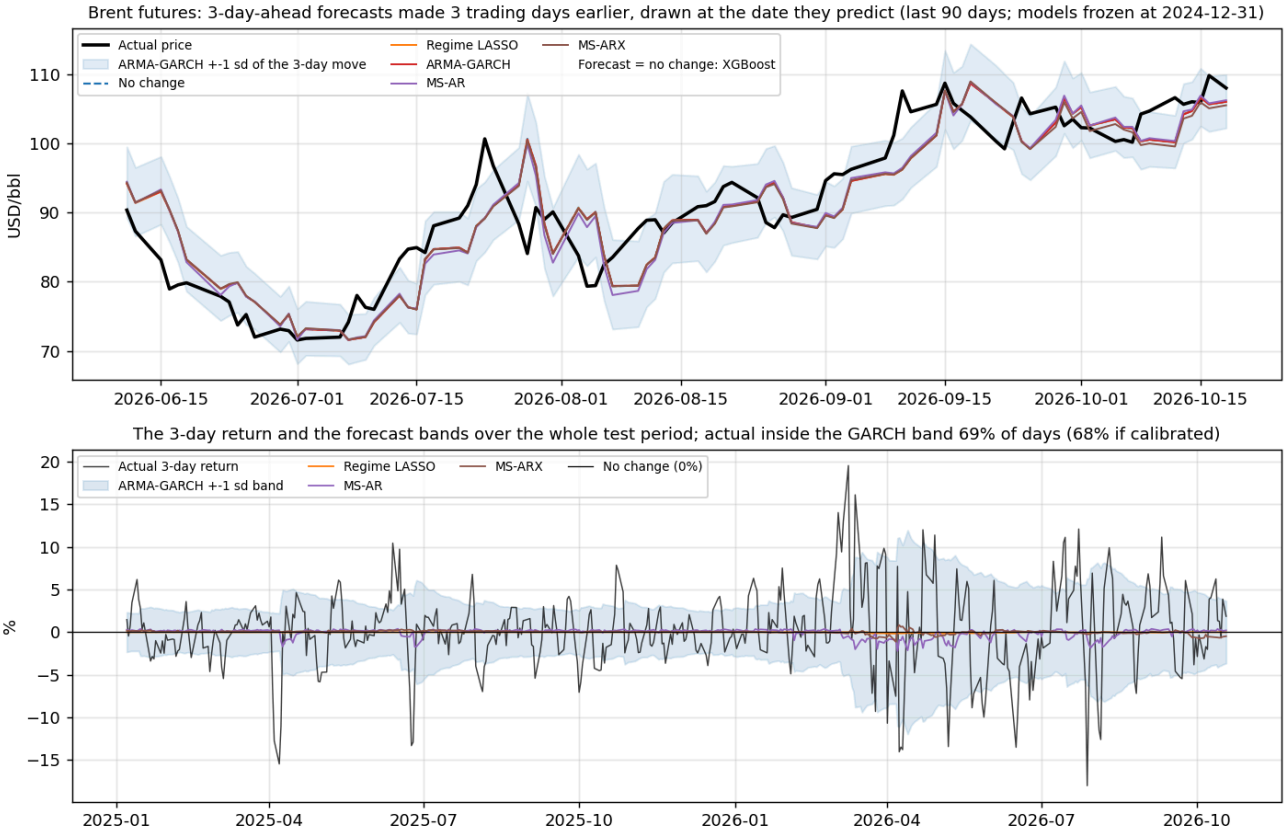
| | | | | | | |
|---|---|---|---|---|---|---|
| +40d | n/a | n/a | n/a | n/a | n/a | n/a |

# Frozen-model check

Models **estimated only on data up to 2024-12-31** (no refitting since), then applied day by day from 2025-01-02 to 2026-10-18 (452 forecast dates). Window and weight choices come from the notebooks' own 2022-2023 validation; nothing after the cutoff was used to fit or choose. This is a deliberately stale test model, not the production model above, so it measures skill plus ageing. Updated monthly (last run 2026-10-19).

## Brent futures

Rolling 3-day-ahead forecast made every day by the frozen models, against what happened:




Accuracy: RMSE of the cumulative return divided by the RMSE of "no change" (below 1 = better; * = Diebold-Mariano p < 0.10; grey = same as no change):

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | Historical mean | AR (direct) | n |
|---|---|---|---|---|---|---|---|---|
| +1d | 1.000 | 1.000 | 1.000 | 1.004 | 1.000 | 1.000 | 1.000 | 451 |
| +2d | 1.000 | 1.000 | 1.000 | 1.008 | 1.004 | 1.000 | 1.000 | 450 |
| +3d | 0.999 | 1.000 | 0.999 | 1.012 | 1.006 | 1.000 | 0.999 | 449 |
| +5d | 0.999 | 1.000 | 1.000 | 1.014 | 1.005 | 1.000 | 1.000 | 447 |
| +10d | 1.000 | 1.000 | 1.000 | 1.020 | 1.003 | 1.000 | 1.000 | 442 |
| +20d | 1.000 | 1.001 | 1.000 | 1.011 | 1.002 | 1.000 | 1.000 | 432 |
| +40d | 1.000 | 1.014 | 1.000 | 0.996 | 1.002 | 1.000 | 1.001 | 412 |
| +60d | 1.000 | 1.011 | 1.000 | 0.979 | 0.997 | 1.001 | 1.001 | 392 |

Direction correct ("no call" = the model forecast exactly no change):

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | Historical mean | AR (direct) |
|---|---|---|---|---|---|---|---|
| +1d | 50% | 54% | 51% | 48% | 46% | 54% | 50% |
| +2d | no call | 45% | 48% | 49% | 44% | 52% | 47% |
| +3d | 50% | 51% | 49% | 48% | 44% | 51% | 51% |
| +5d | 50% | 45% | 48% | 48% | 43% | 51% | 53% |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| +10d | no call | 45% | 51% | 51% | 46% | 52% | 54% |
| +20d | 54% | 49% | 53% | 48% | 43% | 49% | 49% |
| +40d | 58% | 50% | 50% | 44% | 36% | 54% | 50% |
| +60d | no call | 51% | 52% | 55% | 46% | 52% | 49% |

Price error: mean absolute percentage error (%); no change for reference: +1d 1.99%, +2d 2.86%, +3d 3.54%, +5d 4.57%, +10d 6.42%, +20d 8.50%, +40d 12.66%, +60d 15.78%:

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | Historical mean | AR (direct) |
|---|---|---|---|---|---|---|---|
| +1d | 1.99 | 1.99 | 1.99 | 2.01 | 2.00 | 1.99 | 1.99 |
| +2d | 2.86 | 2.86 | 2.86 | 2.90 | 2.87 | 2.86 | 2.86 |
| +3d | 3.54 | 3.54 | 3.54 | 3.58 | 3.56 | 3.54 | 3.54 |
| +5d | 4.56 | 4.57 | 4.57 | 4.63 | 4.59 | 4.57 | 4.57 |
| +10d | 6.42 | 6.42 | 6.41 | 6.55 | 6.42 | 6.42 | 6.42 |
| +20d | 8.50 | 8.50 | 8.50 | 8.68 | 8.51 | 8.50 | 8.50 |
| +40d | 12.66 | 12.76 | 12.66 | 12.89 | 13.08 | 12.65 | 12.68 |
| +60d | 15.78 | 16.05 | 15.78 | 15.56 | 15.90 | 15.76 | 15.77 |

Up-move F1 score:

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | Historical mean | AR (direct) |
|---|---|---|---|---|---|---|---|
| +1d | 0.28 | 0.67 | 0.52 | 0.55 | 0.53 | 0.70 | 0.51 |
| +2d | no call | 0.15 | 0.48 | 0.58 | 0.53 | 0.68 | 0.48 |
| +3d | 0.07 | 0.66 | 0.49 | 0.58 | 0.54 | 0.67 | 0.52 |
| +5d | 0.08 | 0.24 | 0.48 | 0.59 | 0.53 | 0.67 | 0.67 |
| +10d | no call | 0.31 | 0.51 | 0.61 | 0.56 | 0.68 | 0.66 |
| +20d | 0.12 | 0.41 | 0.52 | 0.58 | 0.53 | 0.66 | 0.55 |
| +40d | 0.14 | 0.52 | 0.47 | 0.54 | 0.46 | no call | 0.50 |
| +60d | no call | 0.53 | 0.50 | 0.64 | 0.57 | no call | 0.44 |

Down-move F1 score:

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | Historical mean | AR (direct) |
|---|---|---|---|---|---|---|---|
| +1d | 0.62 | 0.21 | 0.50 | 0.38 | 0.37 | no call | 0.50 |
| +2d | no call | 0.60 | 0.47 | 0.35 | 0.30 | no call | 0.46 |
| +3d | 0.66 | 0.10 | 0.50 | 0.31 | 0.31 | no call | 0.50 |
| +5d | 0.66 | 0.57 | 0.49 | 0.30 | 0.28 | no call | 0.22 |
| +10d | no call | 0.55 | 0.51 | 0.33 | 0.31 | no call | 0.27 |
| +20d | 0.69 | 0.56 | 0.54 | 0.31 | 0.28 | no call | 0.41 |
| +40d | 0.72 | 0.48 | 0.53 | 0.30 | 0.20 | 0.70 | 0.51 |
| +60d | no call | 0.49 | 0.53 | 0.39 | 0.28 | 0.68 | 0.53 |

Share of outcomes inside the +-1 sd band (68% if well calibrated):

| Horizon | Recent volatility | ARMA-GARCH | MS-AR | MS-ARX |
|---|---|---|---|---|
| +1d | 70% | 72% | 76% | 73% |
| +2d | 69% | 72% | 78% | 68% |
| +3d | 67% | 69% | 76% | 67% |
| +5d | 68% | 70% | 76% | 67% |
| +10d | 67% | 71% | 80% | 68% |
| +20d | 69% | 71% | 83% | 71% |
| +40d | 65% | 71% | 81% | 64% |
| +60d | 65% | 70% | 77% | 66% |

Economic value: long or short on the +3-day forecast (flat if it is within 0.1 sd of zero), 5 bp per unit traded, same dates:

| Strategy | Annualised return | Volatility | Sharpe | Max drawdown | Days in market | Costs paid |
|---|---|---|---|---|---|---|
| Regime LASSO | never trades | | | | | |
| XGBoost | never trades | | | | | |
| ARMA-GARCH | never trades | | | | | |
| MS-AR | -10.3% | 14% | -0.70 | -25% | 16% | 0.87% |
| MS-ARX | -14.5% | 14% | -1.01 | -29% | 8% | 0.38% |
| Buy & hold | +21.8% | 45% | 0.66 | -40% | 100% | 0.05% |

Sharpe ratio of the same rule at other horizons:

| Sharpe ratio | +1d | +3d | +5d | +20d | +60d |
|---|---|---|---|---|---|
| Regime LASSO | flat | flat | flat | flat | flat |
| XGBoost | flat | flat | flat | flat | 1.02 |
| ARMA-GARCH | flat | flat | flat | flat | flat |
| MS-AR | 0.71 | -0.70 | -0.52 | -0.31 | 0.11 |
| MS-ARX | 0.12 | -1.01 | -0.36 | 0.19 | -0.13 |
| Buy & hold | 0.66 | 0.66 | 0.66 | 0.66 | 0.66 |

**Robustness**: RMSE / RMSE of no change in sub-periods of the test (below 1 = better than no change). Volatility regime is judged at the forecast date; windows are four consecutive equal slices of the test period.

Horizon +1d

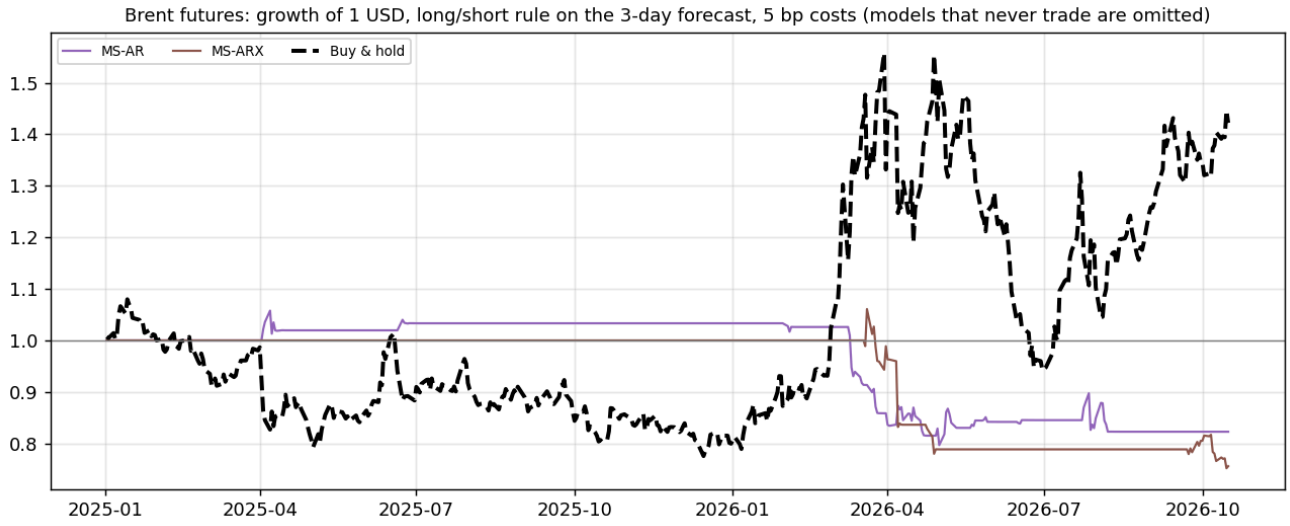
| Model | High vol | Low vol | 2025 | 2026 | Worst window | Windows better |
|---|---|---|---|---|---|---|
| Regime LASSO | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0 of 4 |
| XGBoost | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 3 of 4 |
| ARMA-GARCH | 1.000 | 1.000 | 1.001 | 1.000 | 1.004 | 2 of 4 |
| MS-AR | 1.005 | 1.001 | 0.996 | 1.006 | 1.017 | 3 of 4 |
| MS-ARX | 1.000 | 1.000 | 0.999 | 1.000 | 1.001 | 2 of 4 |
| Historical mean | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 2 of 4 |
| AR (direct) | 1.000 | 1.000 | 1.001 | 0.999 | 1.003 | 1 of 4 |

Horizon +5d

| Model | High vol | Low vol | 2025 | 2026 | Worst window | Windows better |
|---|---|---|---|---|---|---|
| Regime LASSO | 0.998 | 1.000 | 1.000 | 0.998 | 1.000 | 4 of 4 |
| XGBoost | 1.000 | 1.000 | 1.000 | 1.000 | 1.001 | 1 of 4 |
| ARMA-GARCH | 1.000 | 1.000 | 1.000 | 1.000 | 1.001 | 1 of 4 |
| MS-AR | 1.017 | 1.004 | 1.005 | 1.018 | 1.030 | 1 of 4 |
| MS-ARX | 1.008 | 0.996 | 1.001 | 1.007 | 1.012 | 1 of 4 |
| Historical mean | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 2 of 4 |
| AR (direct) | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 3 of 4 |

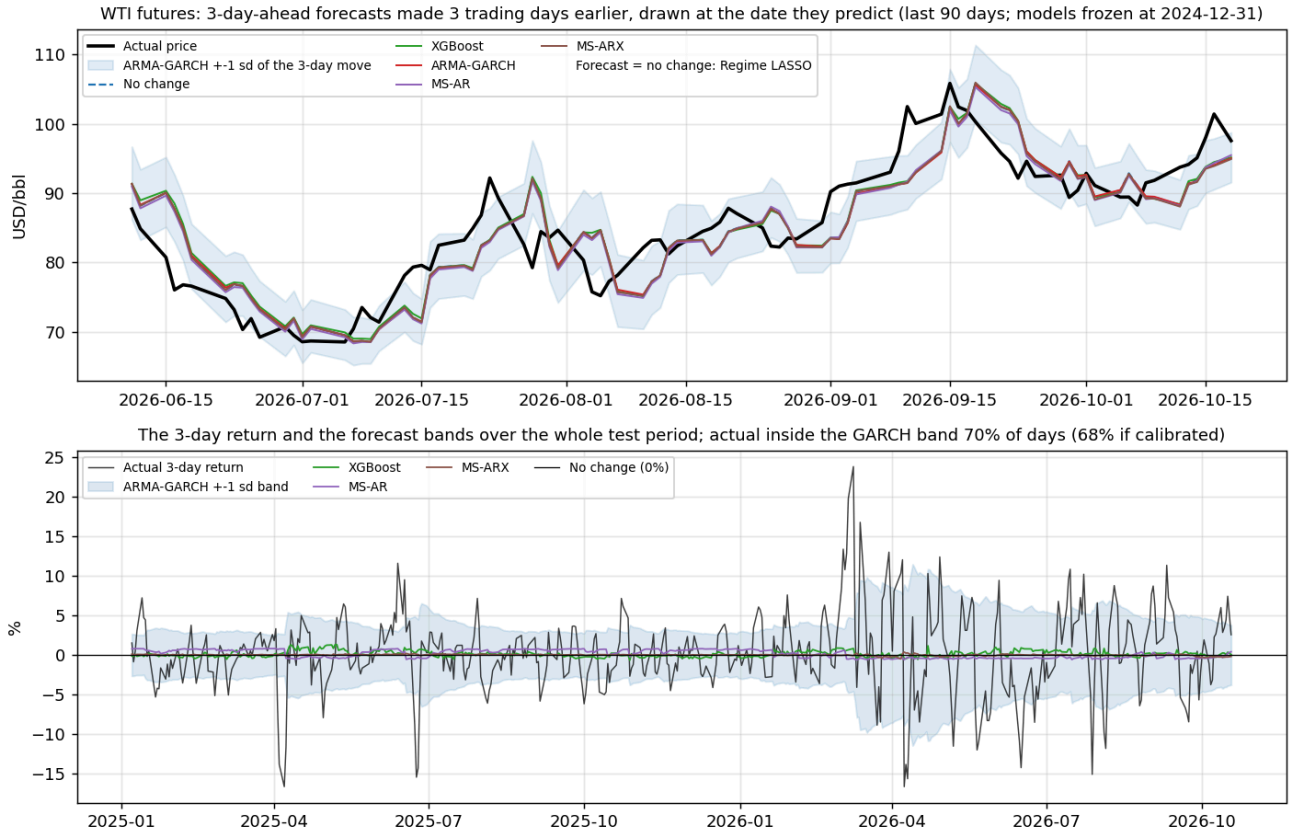
Horizon +20d

| Model | High vol | Low vol | 2025 | 2026 | Worst window | Windows better |
|---|---|---|---|---|---|---|
| Regime LASSO | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1 of 4 |
| XGBoost | 1.001 | 0.997 | 0.997 | 1.001 | 1.003 | 1 of 4 |
| ARMA-GARCH | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 3 of 4 |
| MS-AR | 1.009 | 1.034 | 1.025 | 1.009 | 1.058 | 0 of 4 |
| MS-ARX | 1.001 | 1.015 | 1.021 | 0.999 | 1.065 | 1 of 4 |
| Historical mean | 1.000 | 1.000 | 1.000 | 1.000 | 1.001 | 1 of 4 |
| AR (direct) | 1.000 | 1.001 | 1.001 | 1.000 | 1.002 | 1 of 4 |




## WTI futures

Rolling 3-day-ahead forecast made every day by the frozen models, against what happened:




Accuracy: RMSE of the cumulative return divided by the RMSE of "no change" (below 1 = better; * = Diebold-Mariano p < 0.10; grey = same as no change):

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | Historical mean | AR (direct) | n |
|---|---|---|---|---|---|---|---|---|
| +1d | 1.000 | 0.998* | 1.000 | 1.001 | 1.001 | 1.000 | 1.000 | 451 |
| +2d | 1.000 | 0.993* | 0.999 | 1.002 | 1.002* | 1.000 | 1.000 | 450 |
| +3d | 1.000 | 0.992 | 0.999 | 1.003 | 1.004* | 0.999 | 0.998 | 449 |
| +5d | 1.000 | 0.995 | 1.001 | 1.004 | 1.005* | 0.999 | 1.009 | 447 |
| +10d | 1.000 | 1.002 | 0.999 | 1.012 | 1.007* | 0.999 | 0.995 | 442 |
| +20d | 1.000 | 1.000 | 1.000 | 1.031 | 1.004 | 0.997 | 0.997 | 432 |
| +40d | 1.000 | 1.000 | 1.000 | 1.023 | 1.004 | 0.992 | 0.992 | 412 |

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| +60d | 1.000 | 1.000 | 1.000 | 0.993 | 1.001 | 0.989 | 0.988 | 392 |

Direction correct ("no call" = the model forecast exactly no change):

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | Historical mean | AR (direct) |
|---|---|---|---|---|---|---|---|
| +1d | 49% | 51% | 51% | 48% | 47% | 53% | 52% |
| +2d | 47% | 59% | 51% | 49% | 45% | 52% | 50% |
| +3d | no call | 58% | 50% | 48% | 43% | 51% | 50% |
| +5d | no call | 56% | 50% | 50% | 41% | 50% | 52% |
| +10d | 54% | 52% | 56% | 48% | 40% | 51% | 55% |
| +20d | no call | 50% | 54% | 43% | 41% | 46% | 46% |
| +40d | 53% | 46% | 50% | 40% | 34% | 44% | 44% |
| +60d | 44% | 43% | 51% | 49% | 45% | 46% | 46% |

Price error: mean absolute percentage error (%); no change for reference: +1d 2.07%, +2d 3.01%, +3d 3.75%, +5d 4.82%, +10d 6.47%, +20d 8.54%, +40d 13.03%, +60d 16.05%:

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | Historical mean | AR (direct) |
|---|---|---|---|---|---|---|---|
| +1d | 2.07 | 2.07 | 2.08 | 2.08 | 2.08 | 2.07 | 2.07 |
| +2d | 3.01 | 2.98 | 3.00 | 3.03 | 3.01 | 3.00 | 3.01 |
| +3d | 3.75 | 3.72 | 3.74 | 3.79 | 3.76 | 3.75 | 3.75 |
| +5d | 4.82 | 4.78 | 4.82 | 4.86 | 4.84 | 4.83 | 4.86 |
| +10d | 6.47 | 6.48 | 6.46 | 6.75 | 6.53 | 6.51 | 6.47 |
| +20d | 8.54 | 8.54 | 8.53 | 9.33 | 8.60 | 8.75 | 8.74 |
| +40d | 13.03 | 13.05 | 13.03 | 14.46 | 13.24 | 13.55 | 13.56 |
| +60d | 16.05 | 16.08 | 16.05 | 17.71 | 16.11 | 16.93 | 16.93 |

Up-move F1 score:

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | Historical mean | AR (direct) |
|---|---|---|---|---|---|---|---|
| +1d | 0.59 | 0.56 | 0.50 | 0.52 | 0.50 | 0.69 | 0.69 |
| +2d | 0.58 | 0.62 | 0.51 | 0.52 | 0.49 | 0.69 | 0.62 |
| +3d | no call | 0.61 | 0.50 | 0.51 | 0.48 | 0.67 | 0.62 |
| +5d | no call | 0.59 | 0.49 | 0.52 | 0.48 | 0.66 | 0.57 |
| +10d | 0.58 | 0.59 | 0.56 | 0.51 | 0.50 | 0.67 | 0.64 |
| +20d | no call | 0.64 | 0.51 | 0.46 | 0.50 | 0.63 | 0.63 |
| +40d | 0.57 | 0.57 | 0.46 | 0.45 | 0.43 | 0.61 | 0.61 |
| +60d | 0.30 | 0.60 | 0.48 | 0.58 | 0.55 | 0.63 | 0.63 |

Down-move F1 score:

| Horizon | Regime LASSO | XGBoost | ARMA-GARCH | MS-AR | MS-ARX | Historical mean | AR (direct) |
|---|---|---|---|---|---|---|---|
| +1d | 0.31 | 0.47 | 0.52 | 0.44 | 0.44 | no call | no call |
| +2d | 0.29 | 0.54 | 0.50 | 0.45 | 0.40 | no call | 0.28 |
| +3d | no call | 0.54 | 0.51 | 0.45 | 0.36 | no call | 0.30 |
| +5d | no call | 0.52 | 0.51 | 0.47 | 0.32 | no call | 0.47 |
| +10d | 0.49 | 0.43 | 0.57 | 0.44 | 0.27 | no call | 0.41 |
| +20d | no call | 0.14 | 0.56 | 0.40 | 0.28 | no call | 0.01 |
| +40d | 0.49 | 0.27 | 0.54 | 0.34 | 0.20 | no call | no call |
| +60d | 0.53 | no call | 0.53 | 0.33 | 0.29 | no call | no call |

Share of outcomes inside the +-1 sd band (68% if well calibrated):

| Horizon | Recent volatility | ARMA-GARCH | MS-AR | MS-ARX |
|---|---|---|---|---|
| +1d | 71% | 72% | 67% | 72% |

| | | | | |
|---|---|---|---|---|
| +2d | 69% | 70% | 67% | 73% |
| +3d | 69% | 70% | 66% | 72% |
| +5d | 69% | 68% | 64% | 72% |
| +10d | 69% | 69% | 68% | 76% |
| +20d | 70% | 75% | 69% | 78% |
| +40d | 69% | 70% | 67% | 77% |
| +60d | 65% | 71% | 64% | 72% |

Economic value: long or short on the +3-day forecast (flat if it is within 0.1 sd of zero), 5 bp per unit traded, same dates:

| Strategy | Annualised return | Volatility | Sharpe | Max drawdown | Days in market | Costs paid |
|---|---|---|---|---|---|---|
| Regime LASSO | never trades | | | | | |
| XGBoost | +23.1% | 20% | 1.16 | -17% | 45% | 2.23% |
| ARMA-GARCH | +0.1% | 1% | 0.09 | -1% | 1% | 0.03% |
| MS-AR | +0.0% | 25% | 0.13 | -22% | 62% | 1.68% |
| MS-ARX | never trades | | | | | |
| Buy & hold | +17.4% | 47% | 0.58 | -39% | 100% | 0.05% |

Sharpe ratio of the same rule at other horizons:

| Sharpe ratio | +1d | +3d | +5d | +20d | +60d |
|---|---|---|---|---|---|
| Regime LASSO | flat | flat | flat | flat | flat |
| XGBoost | flat | 1.16 | 1.29 | flat | flat |
| ARMA-GARCH | -0.76 | 0.09 | flat | flat | flat |
| MS-AR | -0.04 | 0.13 | 0.33 | -0.11 | 0.64 |
| MS-ARX | flat | flat | flat | -0.53 | -0.29 |
| Buy & hold | 0.58 | 0.58 | 0.58 | 0.58 | 0.58 |

**Robustness**: RMSE / RMSE of no change in sub-periods of the test (below 1 = better than no change). Volatility regime is judged at the forecast date; windows are four consecutive equal slices of the test period.

Horizon +1d

| Model | High vol | Low vol | 2025 | 2026 | Worst window | Windows better |
|---|---|---|---|---|---|---|
| Regime LASSO | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0 of 4 |
| XGBoost | 0.998 | 0.996 | 0.998 | 0.998 | 1.001 | 3 of 4 |
| ARMA-GARCH | 1.000 | 1.002 | 1.004 | 0.999 | 1.005 | 1 of 4 |
| MS-AR | 1.000 | 1.007 | 1.004 | 1.000 | 1.008 | 2 of 4 |
| MS-ARX | 1.001 | 1.000 | 1.000 | 1.001 | 1.002 | 2 of 4 |
| Historical mean | 1.000 | 1.000 | 1.002 | 0.999 | 1.003 | 1 of 4 |
| AR (direct) | 1.000 | 1.000 | 1.002 | 0.999 | 1.003 | 2 of 4 |

Horizon +5d

| Model | High vol | Low vol | 2025 | 2026 | Worst window | Windows better |
|---|---|---|---|---|---|---|
| Regime LASSO | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0 of 4 |
| XGBoost | 0.996 | 0.987 | 0.994 | 0.995 | 1.014 | 2 of 4 |
| ARMA-GARCH | 1.002 | 0.999 | 0.999 | 1.002 | 1.003 | 1 of 4 |
| MS-AR | 0.996 | 1.032 | 1.027 | 0.995 | 1.038 | 1 of 4 |
| MS-ARX | 1.007 | 0.999 | 1.001 | 1.007 | 1.009 | 1 of 4 |
| Historical mean | 0.998 | 1.004 | 1.008 | 0.995 | 1.018 | 1 of 4 |
| AR (direct) | 1.008 | 1.016 | 0.978 | 1.021 | 1.035 | 1 of 4 |

Horizon +20d

| Model | High vol | Low vol | 2025 | 2026 | Worst window | Windows better |
|---|---|---|---|---|---|---|

| | | | | | | |
|---|---|---|---|---|---|---|
| Regime LASSO | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 0 of 4 |
| XGBoost | 1.000 | 0.999 | 0.999 | 1.000 | 1.002 | 2 of 4 |
| ARMA-GARCH | 1.000 | 0.998 | 0.998 | 1.000 | 1.002 | 3 of 4 |
| MS-AR | 0.994 | 1.284 | 1.230 | 0.990 | 1.383 | 2 of 4 |
| MS-ARX | 1.003 | 1.010 | 1.009 | 1.003 | 1.035 | 0 of 4 |
| Historical mean | 0.988 | 1.068 | 1.054 | 0.986 | 1.129 | 1 of 4 |
| AR (direct) | 0.988 | 1.067 | 1.053 | 0.987 | 1.124 | 1 of 4 |

# Market data, last 7 trading days (USD/bbl)

| Date | WTI futures | WTI spot | Brent futures | Brent spot |
|---|---|---|---|---|
| 2026-10-25 | 101.74 | n/a | 110.40 | n/a |
| 2026-10-23 | 103.27 | n/a | 116.72 | n/a |
| 2026-10-22 | 99.47 | n/a | 113.17 | n/a |
| 2026-10-21 | 97.15 | n/a | 108.82 | n/a |
| 2026-10-20 | 101.79 | 107.89 | 108.79 | 115.22 |
| 2026-10-19 | 103.66 | 110.01 | 109.55 | 118.82 |
| 2026-10-18 | 97.56 | n/a | 108.05 | n/a |

n/a = FRED/EIA spot prices are published with a delay of several days.

Forecasts are expected log-returns turned into prices. "No change" is the random-walk benchmark; futures series are continuous front-month (contract rolls appear as small jumps). The ARMA-GARCH model forecasts almost no change; its value is the volatility band.

Models last trained on data through: arma_garch (2026-10-16), lasso_regime (2026-10-18), msar (2026-10-16), xgboost (2026-10-18). They are retrained quarterly; each day only the new data is fed through.

EIA's futures-curve files stopped in April 2024, so the curve-slope feature is unavailable.